Effects of Greenhouse Gases

Exam code: 0620 & 0971

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Effects of greenhouse gases

Extended Tier Only

- The Sun emits energy in the form of radiation that enters the Earth's atmosphere
- Some of this energy is **reflected** from the Earth's surface
- The Earth's surface **absorbs** energy and then **re-emits** it as thermal energy (infrared radiation)
- Some energy passes through the atmosphere and is **emitted** into space
- Greenhouse gases, such as **carbon dioxide** and **methane**, absorb some of the thermal energy
- These greenhouse gases then re-emit thermal energy in all directions
- This reduces the thermal energy lost to space and traps more thermal energy in the Earth's atmosphere, keeping the Earth warm
  - This process is known as the **greenhouse effect**

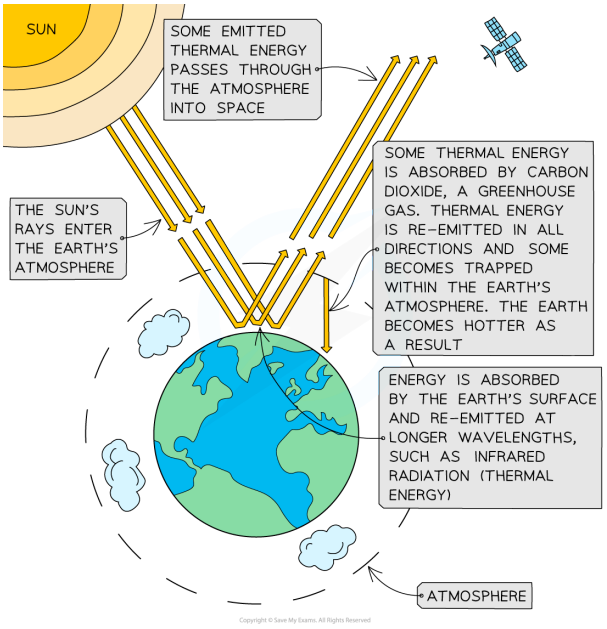

Diagram showing how the greenhouse effect occurs

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- This causes the Earth's average temperature to rise, which is called **global warming**
  - This process is called the **enhanced greenhouse effect**

Consequences of global warming

- Climate change due to the increase in Earth's temperature
- Water levels will rise as glaciers melt because of high temperatures, causing flooding in low-lying countries
- Extinction of species due to the destruction of natural habitats
- Migration of species as they will move to areas that are more habitable (no droughts)
- Spread of diseases caused by warmer climate
- Loss of habitat due to climate change (animals that live on glaciers or in low-lying countries)

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